

University of Asia Pacific
Department of Computer Science and Engineering
CSE 316: Microprocessors and Microcontrollers Lab

CEP Mapping

Project: Voice-Controlled Obstacle Avoiding Car

Submitted By:

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1. Knowledge Profile (K's) Mapping

K's Attributes	How K's Are Addressed Through Our Project	CLO	PLO
K3 – Engineering Fundamentals	Proficiency in Arduino Uno programming, GPIO configuration, PWM-based motor speed control via L298N, and ultrasonic distance measurement using the HC-SR04 sensor, all grounded in core embedded systems engineering fundamentals. (Ref: Section III – System Design; motor and sensor initialization in <code>setup()</code>)	CLO1, CLO2	PLO a, PLO c

K's Attributes	How K's Are Addressed Through Our Project	CLO	PLO
K4 – Specialist Knowledge	Specialist knowledge of HC-05 Bluetooth serial communication, dual-servo scanning (front and rear), median-filter noise reduction (<code>getDistance()</code>), adaptive PWM-based gentle turning, and real-time obstacle avoidance logic with dynamic threshold tuning (<code>safeDistance</code> , <code>clearDistance</code>). (Ref: Section III.C – Bluetooth Control; <code>scanAllDirections()</code>)	CLO2, CLO3	PLO a, PLO c
K5 – Engineering Design	Designing a modular robot architecture with three interdependent sub-systems: (1) Sensing – dual HC-SR04 with servo mounts, (2) Actuation – L298N four-motor drive with speed levels 1–5, (3) Communication – HC-05 Bluetooth with command-driven state machine. Obstacle avoidance integrates quick-look and full-scan strategies. (Ref: Section III.A – System Architecture; <code>avoidObstacleForward()</code> , <code>avoidObstacleBackward()</code>)	CLO3, CLO5	PLO c, PLO e
K6 – Engineering Practice	Implemented a fully functional Bluetooth-controlled obstacle-avoiding robot following embedded systems best practices: non-blocking timed sensor polling (<code>millis()</code> throttle), bubble-sort median filtering, PWM actuation, and structured serial logging for debugging. Hardware assembled and validated on physical prototype. (Ref: Section IV – Implementation; <code>loop()</code> throttle logic)	CLO4, CLO5	PLO c, PLO i
K7 – Comprehension	The system addresses real-world social relevance by providing a low-cost autonomous navigation solution applicable to assistive robotics, home automation, and educational platforms. Safety mechanisms reflect engineering responsibility. (Ref: Section V – Conclusion and Future Work)	CLO6	PLO f, PLO g

2. Complex Engineering Problems (P's) Mapping

P's Attributes	How P's Are Addressed Through Our Project	CLO	PLO
P1 – Depth of Knowledge Required	Resolving obstacle avoidance in real-time required in-depth knowledge of ultrasonic timing physics, PWM motor control, servo PWM calibration, and Bluetooth UART communication, all at the level of K3, K4, and K6. (Ref: Section III.B – Distance Measurement; <code>getDistance()</code> and <code>executeTurn()</code>)	CLO1, CLO2, CLO5	PLO a, PLO c
P2 – Range of Conflicting Requirements	The system must simultaneously satisfy conflicting demands: fast response to obstacles vs. stable median-filtered readings; smooth Bluetooth-commanded motion vs. autonomous override; gentle curved turning vs. hard obstacle avoidance turns. These were resolved through the <code>gentleTurning</code> flag and throttled distance checks. (Ref: <code>loop()</code> – command handling vs. distance check section)	CLO2, CLO3	PLO b, PLO c
P3 – Depth of Analysis Required	No standard solution existed for integrating dual-directional scanning with a Bluetooth override state machine. Abstract reasoning was required to design the quick-look strategy (left/right check before full 5-angle scan) and the iterative backing-up logic when the robot is fully blocked. (Ref: <code>avoidObstacleForward()</code> – Step 1 quick look; Step 2 full scan loop)	CLO3, CLO5	PLO b, PLO e
P7 – Interdependence	The project is a high-level system composed of many interdependent sub-problems: Bluetooth serial parsing, dual-servo position control, dual ultrasonic measurement, four-motor PWM drive, speed-level state management, and obstacle avoidance state transitions, all running in a single cooperative loop. (Ref: <code>setup()</code> initialization sequence; full <code>loop()</code> integration)	CLO1, CLO3, CLO5	PLO a, PLO c, PLO i

3. Complex Engineering Activities (A's) Mapping

A's Attributes	How A's Are Addressed Through Our Project	CLO	PLO
A1 – Range of Resources Used	The project utilises diverse resources including: Arduino Uno, L298N motor driver, two HC-SR04 ultrasonic sensors, HC-05 Bluetooth module, two servo motors, four-wheel chassis with DC motors, Li-ion batteries, Arduino IDE, Bluetooth serial terminal, and structured firmware with multiple libraries (Servo.h). (Ref: Section II – Hardware; Table I – Components)	CLO1, CLO5	PLO a, PLO e
A2 – Level of Interaction	Significant interaction occurs between hardware and software layers: Bluetooth commands alter the robot's motion state; sensor data overrides Bluetooth commands during obstacle events; servo positions affect sensor field-of-view; motor PWM values are dynamically updated by speed commands and avoidance logic. (Ref: <code>loop()</code> – command-state flags <code>goingForward</code> , <code>goingBackward</code> , <code>gentleTurning</code>)	CLO2, CLO3	PLO c, PLO i
A3 – Innovation	The project innovatively combines: (1) dual front-and-rear obstacle detection for bidirectional protection, (2) a gentle-curve turning mode using differential PWM (<code>gentleRight()</code> , <code>gentleLeft()</code>), (3) a 5-level speed control system with proportional <code>gentleFactor</code> scaling, and (4) a median-of-5 noise-filtering algorithm for stable ultrasonic readings. (Ref: <code>getDistance()</code> ; speed cases '1'–'5')	CLO3, CLO5	PLO c, PLO e
A4 – Consequences for Society & Environment	The system promotes accessible, low-cost autonomous navigation applicable to assistive devices, warehouse robotics, and educational STEM platforms. Efficient PWM control reduces power consumption, extending battery life and lowering electronic waste. Real-world deployment can reduce human exposure to hazardous environments. (Ref: Section V – Social Relevance and Future Work)	CLO4, CLO6	PLO f, PLO g

A's Attributes	How A's Are Addressed Through Our Project	CLO	PLO
A5 – Familiarity	The project extended beyond standard lab exercises by integrating bidirectional sensing, Bluetooth-driven state machines, and novel differential-drive gentle turning, applying first-principles embedded design in ways not covered by existing mini-project templates. (Ref: Full firmware; <code>avoidObstacleBackward()</code> ; gentle turn implementation)	CLO3, CLO5	PLO c, PLO 1